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BEL-E

Business Management and Logistics Processes 1

Lecture 1 | Introduction to logistics

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Assessment

Group case study

Details and assessment criteria will be provided later in the module.

Final examination

90 minutes in January, based on material covered during the module.

Keep examples from class: they will help you apply the concepts.

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What this module covers

Business decisions

Strategy, marketing, pricing and procurement

Physical flows

Materials, production, warehousing and distribution

Financial information

Accounting, inventory values and financial statements

Supply chain design

Storage, location choices and forecasting

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By the end of this lecture

- Explain the roles of business management and logistics in bringing a product to customers
- Trace a product from supplier to customer and back
- Apply the 7 Rs to a delivery problem
- Explain why cost, stock, speed and service can conflict

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The Aster One launch

Aster Audio is preparing to launch its first wireless earbuds, Aster One, in Germany.

Inputs

Batteries, chips, cases and packaging

Operations

Assembly, testing and order fulfilment

Customers

Online orders, delivery and returns

Question: What could go wrong before the customer hears the first song?

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Business management and logistics

Business management

Decides what to sell, at what price, to whom, and with which resources.

Shared decisions

Balances customer service, cost, capacity and risk across the business.

Logistics

Plans and manages the flow of goods and the information needed to move them.

Aster One example: a launch date connects sales plans with parts, assembly and delivery.

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Does everyone face scarcity, even a billionaire or a government? What might each lack?

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Scarcity explained

Resources are limited and have alternative uses. Everyone faces choices, whatever their income.

A student

Limited hours to study, work or rest.

A billionaire

Limited time, attention and skilled people.

A government

A budget cannot fund every public priority.

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Choice and opportunity cost

Aster Audio has a limited launch budget. Faster battery deliveries leave less for marketing or testing.

Faster batteries

Reduce the risk of a delayed Aster One launch.

Marketing or testing

Promote Aster One or strengthen quality checks.

What would you prioritise, and what would you give up?

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What is business management?

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What is business management?

Business management is about making decisions with limited resources to achieve an organisation's goals.

Resources

Money, people, time and materials

Decisions

What to make, buy, price and invest in

Results

Customer value and financial performance

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What is logistics?

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What is logistics?

Logistics manages the movement and availability of goods, together with the information that supports those flows.

Place

Move Aster One earbuds from the assembly site to the customer.

Time

Have them available when customers want them.

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Origins of logistics

Logistics originally helped armies keep essential supplies available.

Military use

Administration, storage and control of transport for wartime supplies.

Trade routes

The Silk Road and Salt Road made organised movement of goods important for trade.

Silk Road video: https://www.youtube.com/watch?v=vn3e37VWc0k&t=27s

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How logistics adds value

Place

The product reaches the market where it is wanted.

Time

It arrives when it is useful to the customer.

Condition

It arrives complete, protected and working.

Packaging may protect value; excessive packaging also adds cost and waste.

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The 7 Rs of logistics

Right product

Aster One earbuds ordered

Right quantity

Correct number of units

Right condition

Working and undamaged

Right time

When promised

Right place

Correct destination

Right customer

Correct recipient

Right cost

Delivered at a reasonable cost

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Delivery problem: which Rs failed?

A customer orders two pairs of Aster One earbuds for Friday. One pair arrives on Monday with a damaged charging case.

Identify

Which of the 7 Rs were missed?

Recommend

What should the firm do next?

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Challenges for logistics

Demand changes

A promotion creates more orders than expected.

Supply disruption

Battery deliveries arrive later than planned.

Returns

Customers send faulty or unwanted items back.

Cost and sustainability

Fast delivery and packaging consume resources.

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Why logistics matters

Manufacturers, retailers and customers depend on reliable links between production, trade and transport.

For companies

Late parts can interrupt production or delay sales.

For customers

Availability and delivery shape the buying experience.

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Why logistics matters worldwide

“Maritime transport is the backbone of international trade and the global economy.” — UNCTAD

Over 80%

of world trade in goods by volume is carried by sea.

12.1 billion tonnes

of goods were loaded for international shipping in 2024.

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Why logistics matters in Germany

Germany exported goods worth €1.56 trillion in 2024. Reaching customers abroad depends on logistics.

€335 billion

estimated logistics market volume in Germany in 2024.

3.35 million people

worked in logistics in Germany in 2024.

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Different ways to organise supply

Forecast and hold stock

Build or buy ahead of demand. Stock helps absorb uncertainty but ties up money and space.

Respond to orders

Make or replenish after orders arrive. This can reduce stock but needs reliable suppliers and capacity.

Most firms combine approaches for different products and components.

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The Aster One supply chain

Supplier	→	Inbound	→	Site	→	Outbound	→	Customer
Parts		Deliveries		Assembly		Shipping		Use

Material flow

Components and finished Aster One earbuds move forward.

Information flow

Orders, forecasts and stock data guide decisions.

Payment and other financial flows link customers, firms and suppliers.

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Inbound, in-house and outbound

Inbound

Receive batteries and other parts from suppliers.

In-house

Store, move, assemble and test within the site.

Outbound

Pick, pack and deliver customer orders.

Returns create a reverse flow for inspection, repair, reuse or recycling.

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Intralogistics: inside the site

Intralogistics concerns the movement, storage and related information flows within a facility.

Receive

Unload and check incoming parts.

Supply assembly

Bring the right parts to the workstation.

Dispatch

Move packed orders to shipping.

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Materials management

The firm must know what it has, where it is and when more is needed.

Stock records

Track quantities and locations of batteries, cases and finished Aster One earbuds.

Quality checks

Identify damaged or unsuitable parts.

Replenishment

Order before essential parts run out.

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Purchasing and procurement logistics

Purchasing

Choose suppliers and agree what to buy and at what price.

Procurement logistics

Plan and manage the delivery and receipt of those inputs.

Aster Audio decision: a lower battery price may be unattractive if quality or delivery reliability is poor.

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Production logistics

Production needs the right parts at each workstation when they are required.

Before assembly

Make parts available and schedule work.

During assembly

Move and track items between steps.

After assembly

Test and transfer finished units.

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Warehouse logistics

Why store?

A stock buffer can help meet demand or protect against delayed supply.

What does it cost?

Storage uses space and labour; money remains tied up in inventory.

The goal is an appropriate stock level for the promised service.

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Distribution and returns

To the customer

Pick the order, protect it, choose delivery arrangements and provide tracking.

Back from the customer

Receive returns, inspect them and decide on repair, resale or responsible disposal.

Good service includes what happens after the delivery.

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Logistics goals can conflict

Fast delivery

May require nearby stock and spare capacity.

Low inventory

Releases cash but increases the risk of shortages.

Low cost

May favour consolidated, slower transport.

Managers choose a balance that supports the company's customer promise.

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Decision exercise: stock before launch

Demand for Aster One is uncertain. Batteries take time to replenish.

Option A

Hold extra batteries before launch.

Option B

Order fewer batteries and replenish as sales arrive.

Which option would you choose, and what further information do you need?

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What determines a logistics design?

The product

Value, size, fragility, shelf life and handling needs.

The market

Customer location, order pattern and delivery promise.

Compare Aster One earbuds with fresh food: the same logistics design would not fit both.

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Takeaways

- Business choices and physical flows have to work together.
- Logistics connects suppliers, operations and customers.
- A better service level usually requires a conscious cost and risk decision.